IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

GitHub Repository

- The GitHub repository linked here contains all the completed notebooks and python files.

Executive Summary

- Summary of methodologies
  - Data collection via API and web scraping
  - Data cleaning and preparation (wrangling) including SQL queries and exploratory data analysis
  - Feature engineering and data standardization
  - Development of interactive visualizations – mapped launch sites (Folium) and created dashboard (Plotly)
  - Model building and evaluation – implementing various supervised machine learning algorithms for predictive classification models

- Summary of all results
  - Key factors impacting landing success included launch site and payload mass.
  - Observed similar model performance accuracy with logistic regression, support vector machine (SVM) and K-Nearest Neighbors (KNN).
  - Decision tree model was most accurate with the fewest false-positives.

Introduction

- Project Background and Context:

Rocket launches are expensive endeavors. Compared to competitors, SpaceX manages to reduce costs associated with rocket launches by recovering and reusing the rocket's first stage, when possible. The first stage of a rocket contains the most powerful engines and provides the primary thrust and acceleration. The first stage cannot always be recovered and is sometimes destroyed in-flight.

- Problems of Interest:

This project aims to utilize publicly available data to determine if the first stage of a SpaceX Falcon 9 lands successfully. Understanding if the first stage will successfully land, informs the possibility of first stage reuse and the cost of a launch.

Section 1

Methodology

Methodology

Executive Summary

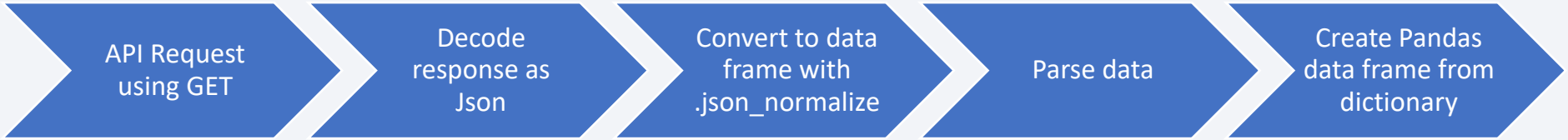
- Data collection methodology:
  - Launch information available from SpaceX REST API and normalized .json data structure into flat table
  - Supplemental information on Falcon 9 launch records available by web-scraping related Wiki pages and parsed HTML data with BeauitfulSoup
- Perform data wrangling
  - Linked data from API and Wiki by ID number, restricted data to information on Falcon 9 launches, addressed null values
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Data was standardized (preprocessing), split into training and testing data, trained model, and performed grid search to identify hyperparameters that performed best for algorithms tested.

Data Collection

- Activities were performed in Jupyter Lab with Python
- SpaceX REST API
  - Required Python libraries: requests, pandas, numpy, datatime
- Web Scraping
  - Required Python libraries: beautifulsoup4, requests, pandas, unicodedata

Data Collection – SpaceX REST API

- Available data included: rocket type, launch site being used with latitude and longitude, mass of payload, orbit, outcome of landing, type of landing, number of flights with core, etc.
- [GitHub Link for Code]

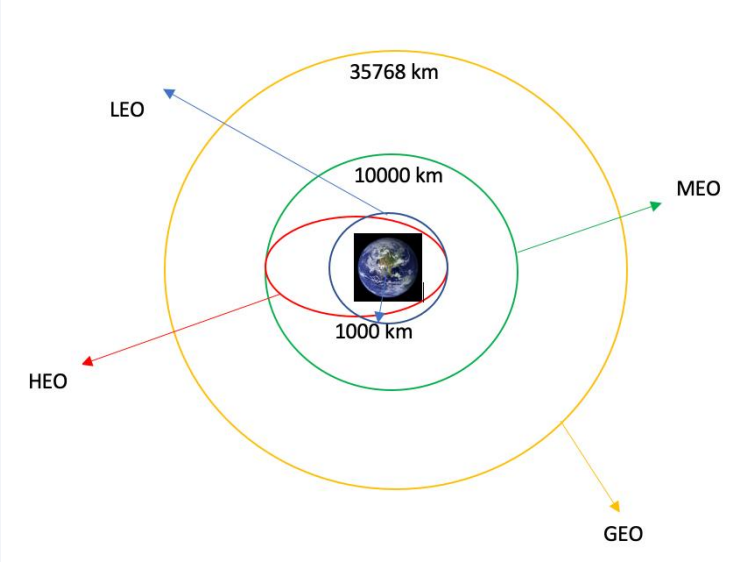


Data Collection - Scraping

- Web scraping from Wikipedia (webpage updated: June 9, 2021)
- Available data include: flight number, date of launch, booster version, landing status, customer, etc.
- GitHub Link for Code

Data Wrangling

- Required Python libraries: pandas, numpy
- Data from SpaceX REST API
  - Filtered to only include Falcon 9 launches
  - Replace missing values of 'PayloadMass' with the mean
  - Understand frequencies with .value_counts()
    - Launches on each site
    - Orbits
    - Mission outcomes
  - Create landing outcome label to classify first stage landing as successful or unsuccessful
- [GitHub Link for Code]



Filtering → Replace Missing Values → Review Frequencies → Create outcome classification variable

EDA with Data Visualization

- Required Python libraries/extensions: piplite, pandas, numpy, seaborn, matplotlib.pyplot, fetch, io
- To understand the correlation between variables, scatter plots were developed and overlayed the outcome of the launch (successful first stage landing [‘Class’=1] or unsuccessful first stage landing [‘Class’=0]). The scatterplots created included:
  - Flight Number vs Payload Mass
  - Flight Number vs Launch Site
  - Payload Mass vs Launch Site
  - Flight Number vs Orbit Type
  - Payload Mass vs Orbit Type
- To understand the success rate of each orbit, we created a bar chart.
- A line chart was used to visualize the annual average success rate.
- [GitHub Link for Code]

EDA with SQL

- Required Python libraries/extensions: sqlalchemy (v.1.3.9), ipython-sql, prettytable, csv, sqlite3, pandas
- GitHub Link for Code

- Established connection with database
- SQL Queries Performed:
  - Remove blank rows from table
  - Names of unique launch sites
  - Records with lunch site that began with string 'CCA'
  - Total payload mass carried by boosters launched by NASA (CRS)
  - Average payload mass carried by booster version F9 v1.1
  - Date with first successful ground pad landing
  - Boosters with success in drone shop and a payload mass between 4,000 and 6,000
  - Total successful and failure mission outcomes
  - Booster versions that carried the maximum payload mass
  - Failure drone ships in 2015
  - Ranking landing outcomes by count that occurred between 2010-06-04 and 2017-03-20

Build an Interactive Map with Folium

- Required Python libraries/extensions: piplite, folium, pandas
- Objects were added to the map understand launch site geography, and the proximity of launch sites with major features such as the coastline, railways, highways, and cities.
- These maps can aid in the identification of an optimal launch site and identify if any patterns that exist, such as which launch sites have relatively high success rates.
- Map objects added:
  - NASA's Johnson Space Center and launch sites were marked with a circle and labelled.
  - Launch outcomes for each site were added as a marker cluster with successful outcomes marked in green and unsuccessful outcomes marked in red.
  - A marker was added to a point on the coastline, and a polyline was drawn between the launch site and coastline point.
  - Distance between the launch site and major objects (highway, railway, airport, city)
  - Mouse position latitude and longitude was added for user convenience
- GitHub Link for Code

Build a Dashboard with Plotly Dash

- Required Python libraries/extensions: pandas, dash, plotly
- The graphics available within this dashboard can aid stakeholders in exploring SpaceX Falcon 9 launch data.
- A pie chart was developed and the user can view success counts for all launch sites, and a scatter plot shows playload mass (kg) and mission outcome by booster version for all sites.
- A range slider allows for the user to select a payload range of interest.
- Additionally, the user can select different launch sites to examine more closely.
  - When a specific launch site is selected, the pie chart illustrates the success and failures for that launch site. The scatter plot also updates to show payload masses and mission outcome, by booster version, for the specific site.
- GitHub Link for Code

Predictive Analysis (Classification)

- Required Python libraries/extensions: piplite, pandas, numpy, matplotlib.pyplot, seaborn, sklearn, fetch, io
- GOAL: Determine the predictive classification model that has the best accuracy for predicting if the first stage of the Falcon 9 lands successfully.
- [GitHub Link for Code]

Predictive Analysis (Classification) cont.

- Data was restricted to variables of interest following exploratory data analysis. Categorical columns were converted to dummy variables utilizing OneHotEncoder .get_dummies() and cast to numeric columns as float64.
- The independent variables (X) were standardized using .StandardScaler() and .fit_transform() was applied.
- The dependent variable of interest for classification 'Class' was assigned as numpy array set to variable (Y). 'Class'=1 when the first stage lands successfully, and 'Class'=0 when the first stage landing was unsuccessful.
- Data was split into training and testing datasets. The parameter test size was set to 0.2 and the random_state was set to 2. This resulted in 18 test samples.
- Multiple model algorithms were used and conduct a grid search cross validation (cv=10) to identify the hyperparameters (best_params_) that provided the best accuracy (best_score_) for a given algorithm.
    - Logistic Regression
    - Support Vector Machines (SVM)
    - Decision Tree Classifier
    - K-nearest Neighbors (KNN)
- We then apply this model to the testing data to identify the accuracy (score) and create a confusion matrix (plot_confusion_matrix).

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

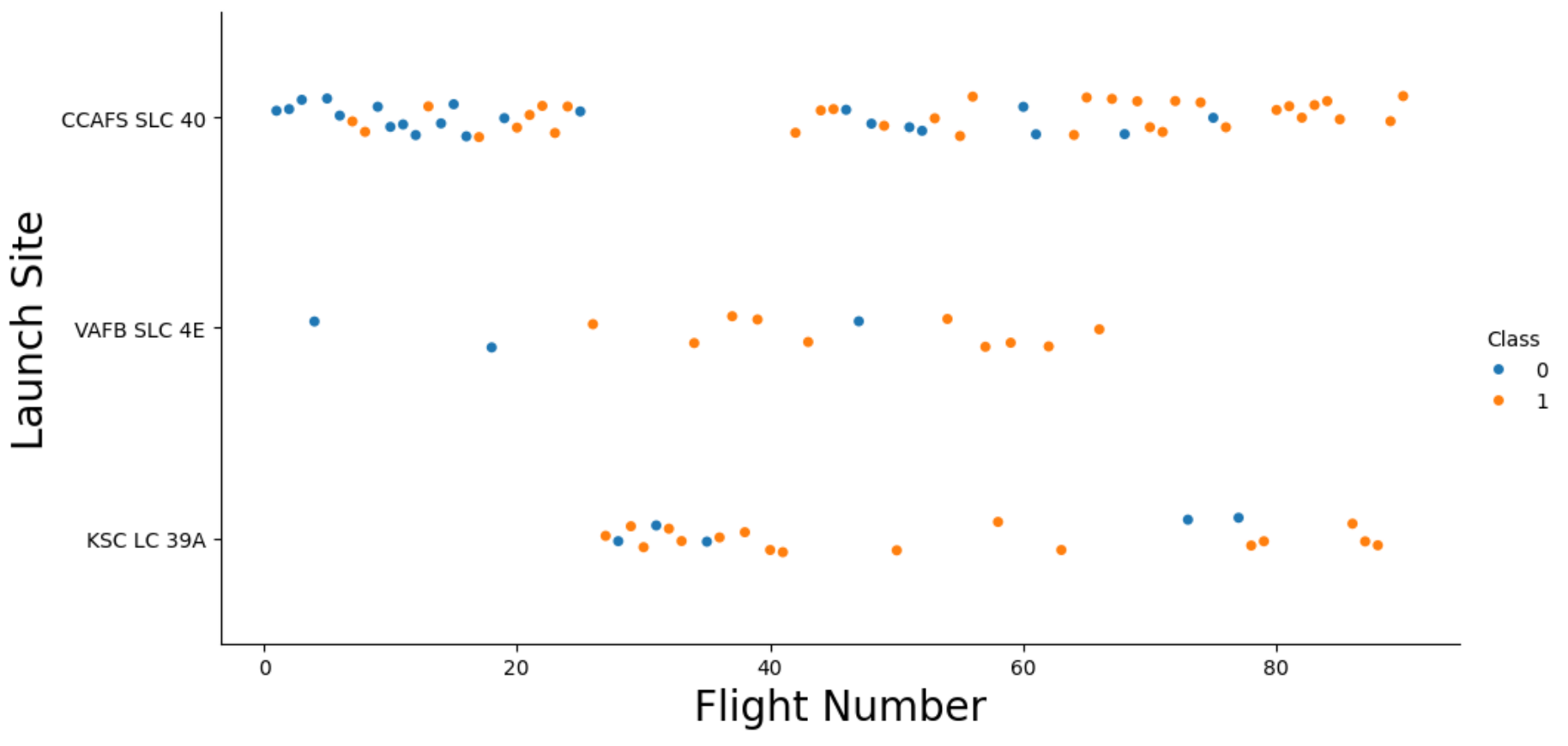
Section 2

Insights drawn from EDA

EDA Results

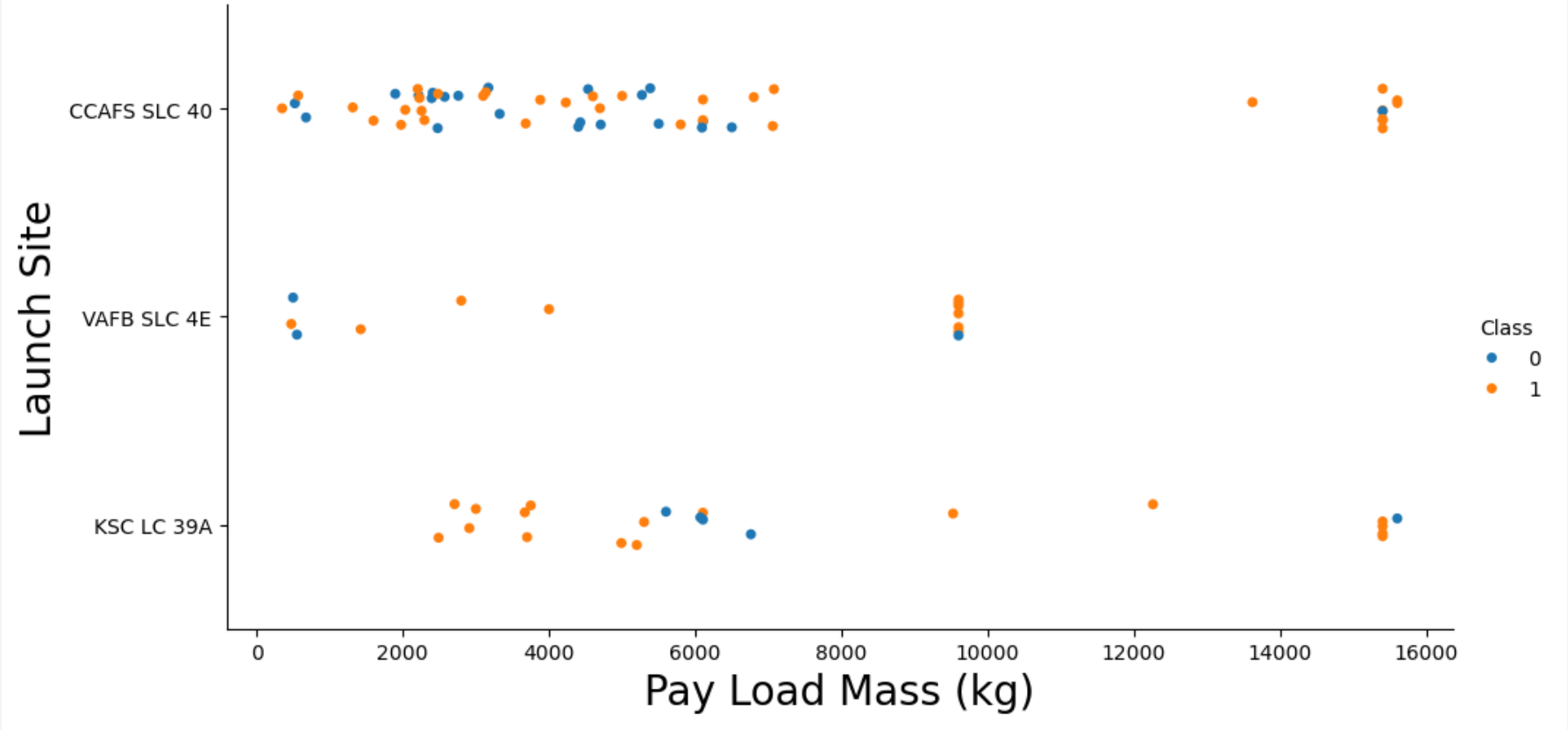
- Exploratory data analysis identified key variables associated with the first stage landing successes that could be used in modeling.
  - 'FlightNumber'
  - 'PayloadMass'
  - 'Flights'
  - 'Block'
  - 'ReusedCount'
  - 'Orbit'
  - 'LaunchSite'
  - 'Serial'
  - 'GridFins'
  - 'Reused'
  - 'Legs'
  - 'LandingPad'

Flight Number vs. Launch Site



We see later flight numbers were associated with more successful first stage landings. Launch site CCAFS SLC 40 had the largest number of launches. Early flights at all sites were not successful.

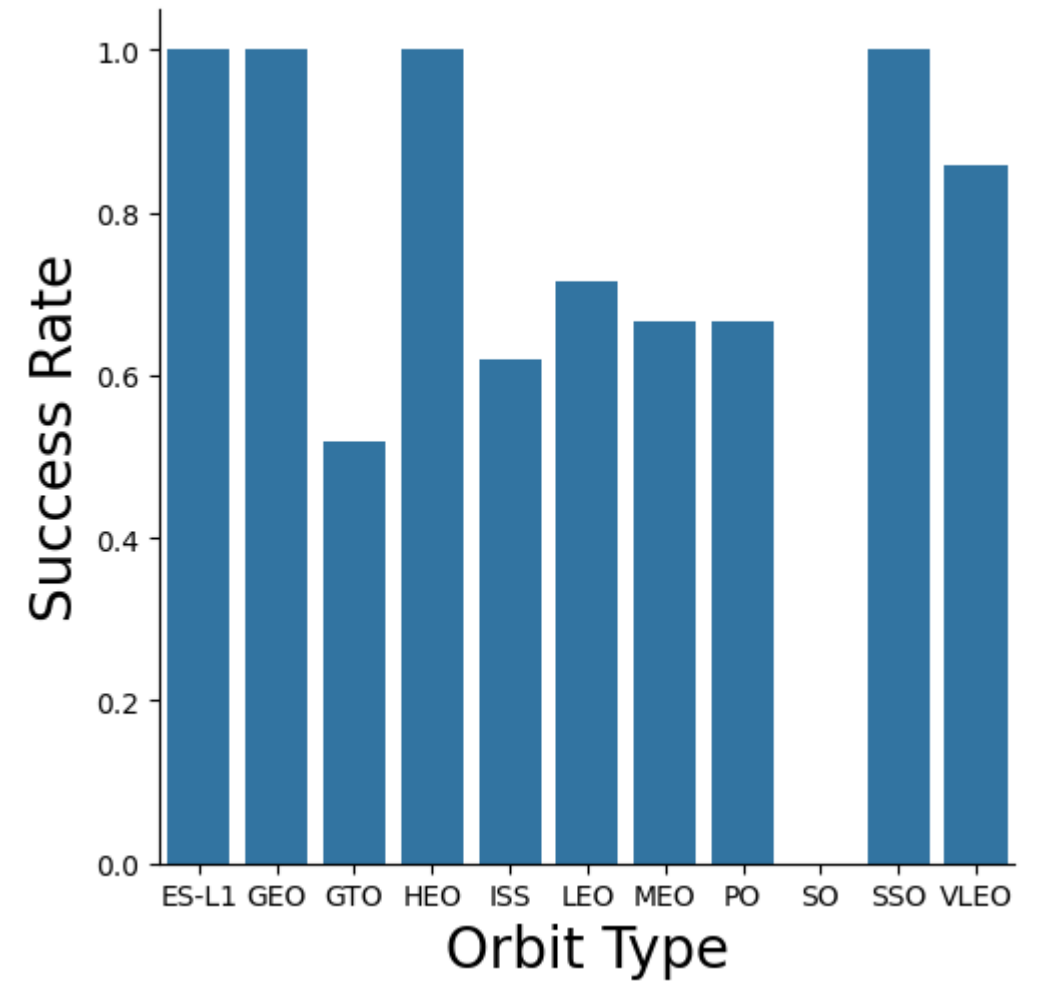
Payload vs. Launch Site



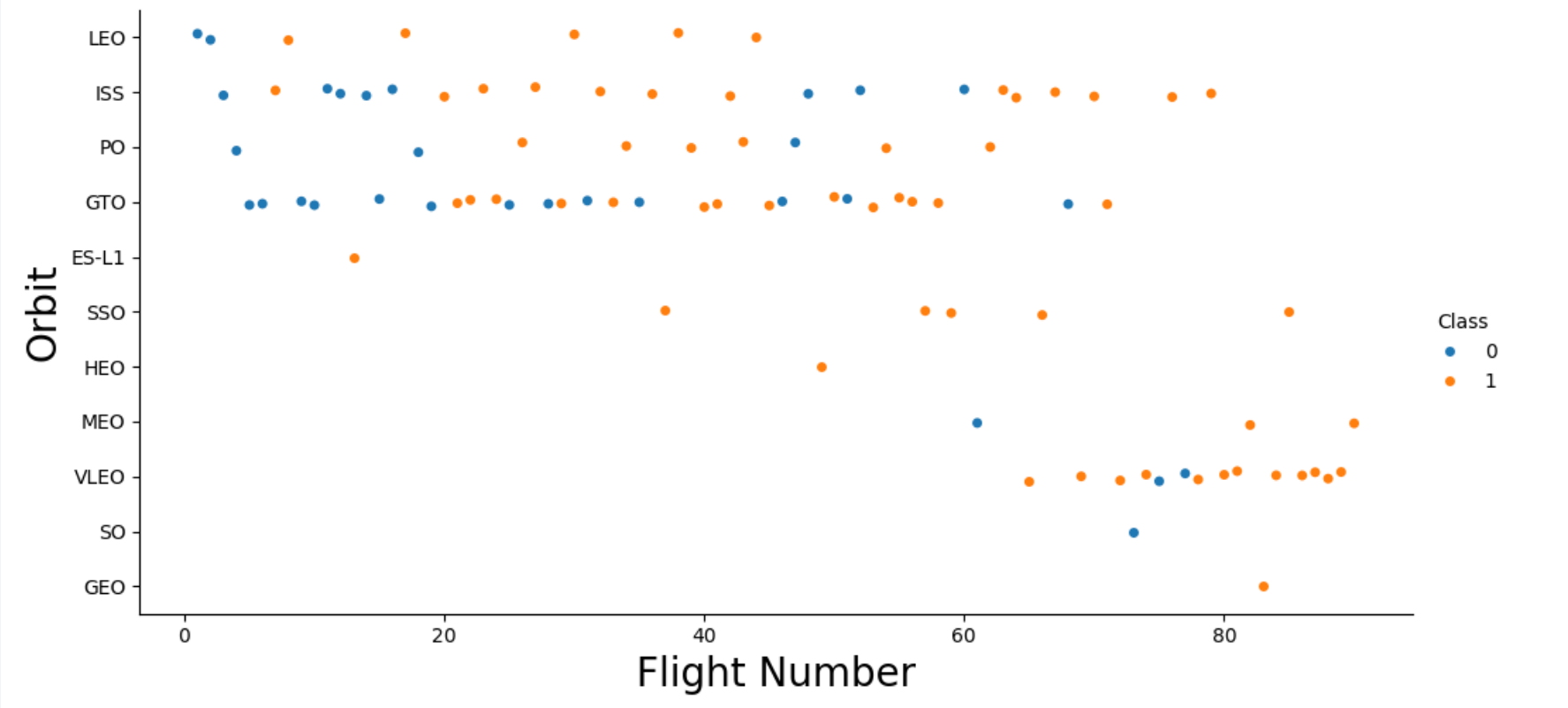
Launch site VAFB SLC 4E never had a payload mass above 10,000 kg, although majority of these were successful. Across all launch sites we see few failures at high payload masses.

Success Rate vs. Orbit Type

From this bar chart we see a 100% success rate among rockets that reached ES-L1, GEO, HEO, and SSO. All orbits, excluding SO, had more than 50% success.

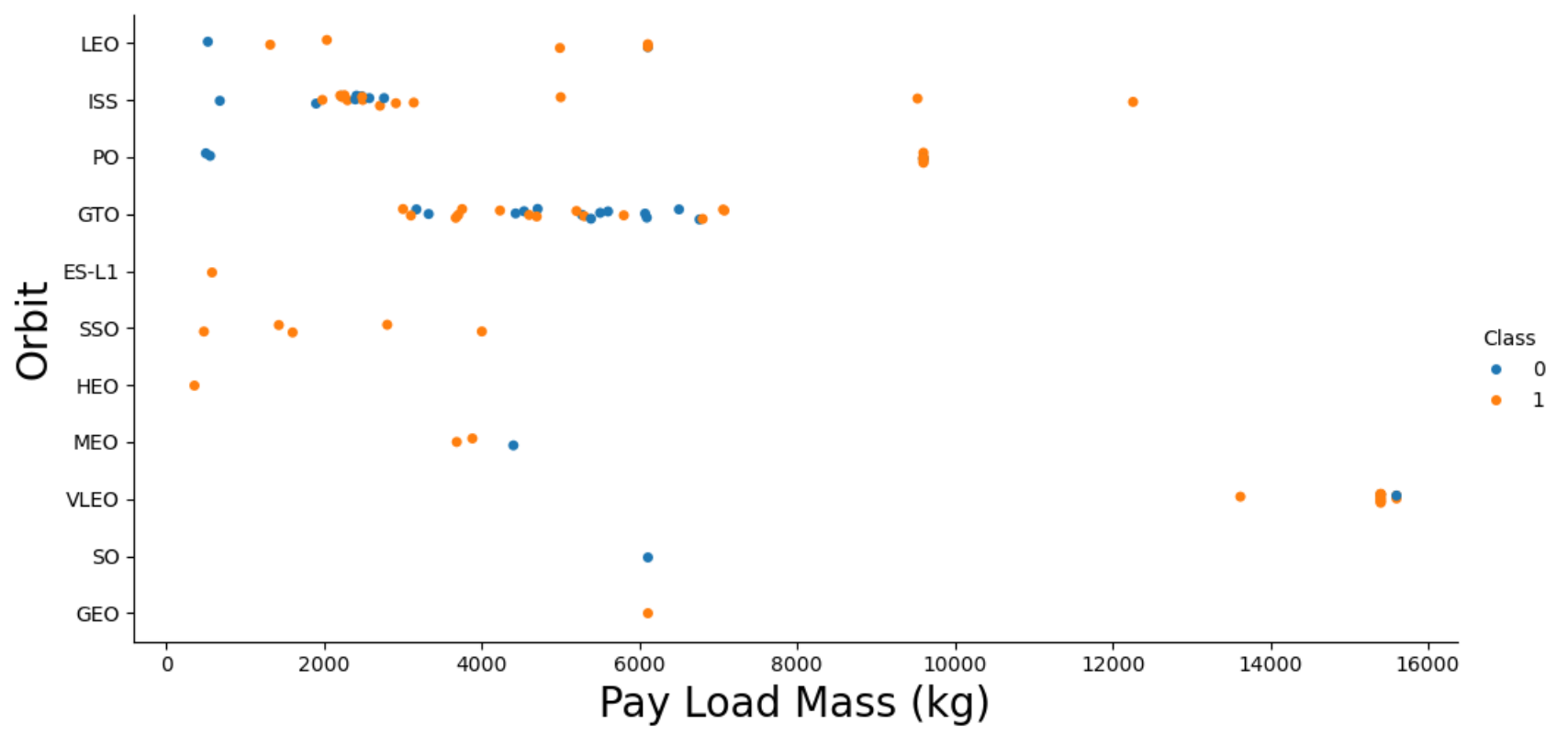


Flight Number vs. Orbit Type



From this scatter plot we can see early flight numbers did not enter SSO, HEO, MEO, BLEO, SO, or GEO orbit. In the LEO orbit, later flight numbers were successful.

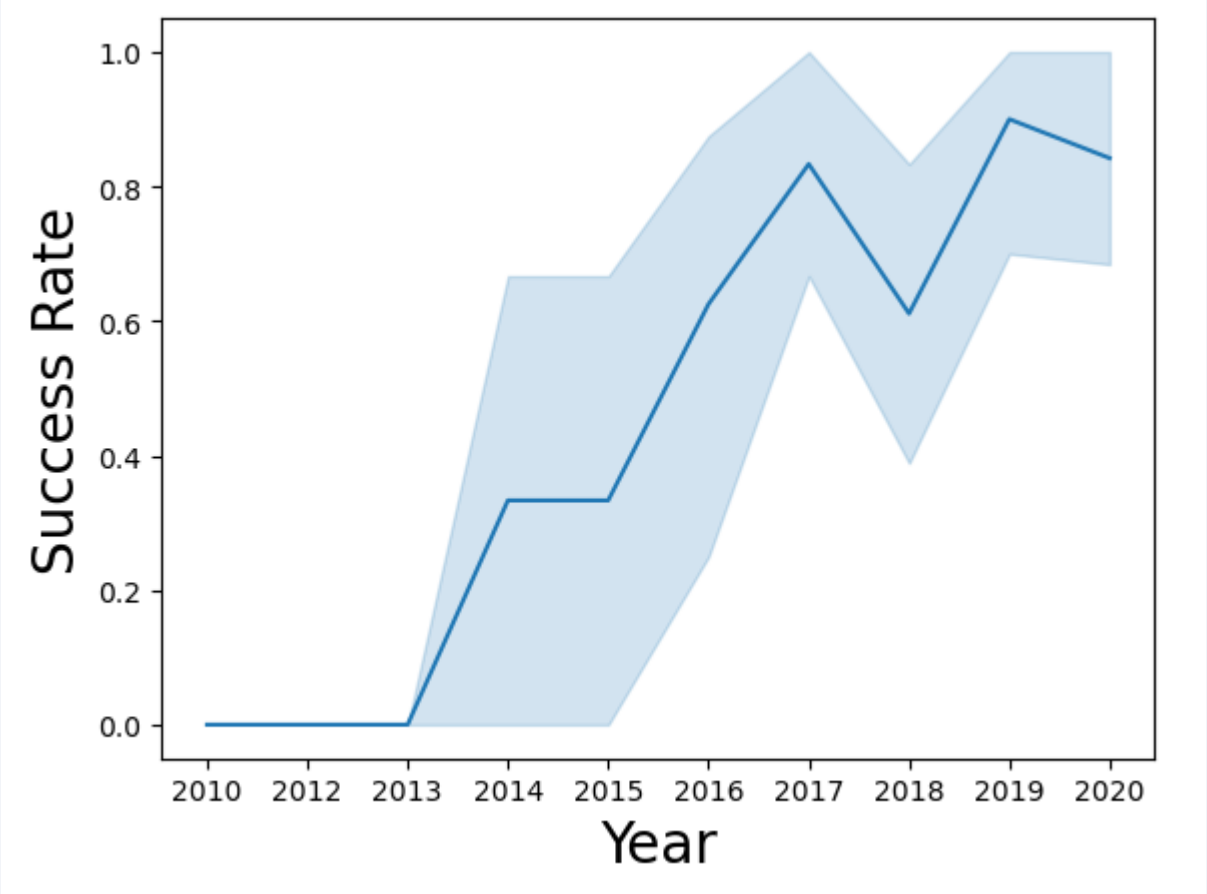
Payload vs. Orbit Type



From this plot we observe the ES-L1, SSO, and HEO orbits had all successes, and payload masses under 5,000kg. Among LEO, ISS, and PO, heavy payload masses were more successful.

Launch Success Yearly Trend

There were no successes prior to 2013. However, since then, the success rate has been increasing. There was a slight decrease in success in 2018. Overall, there appears to be an increase in average annual success rate.



All Launch Site Names

- There are four unique launch sites:
  - CCAFS LC-40
  - CCAFS SLC-40
  - KSC LC-39A
  - VAFB SLC-4E

Launch Site	Lat	Long
CCAFS LC-40	28.562302	-80.577356
CCAFS SLC-40	28.563197	-80.576820
KSC LC-39A	28.573255	-80.646895
VAFB SLC-4E	34.632834	-120.610745

Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" like 'CCA%' limit 5
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- The following 5 records are all from launch site CCAFS LC-40 with two failed landings, and three had no landing attempted. The payloads varied yet all used booster F9 v1.0.

Total Payload Mass

- Boosters launched by NASA (CRS) carried a total payload mass of 45,596kg.

Display the total payload mass carried by boosters launched by NASA (CRS)

```
%sql SELECT sum("PAYLOAD_MASS__KG_") as tot_payload_mass FROM SPACEXTABLE WHERE "Customer"= "NASA (CRS)"
```

```
 * sqlite:///my_data1.db
Done.
```

tot_payload_mass
45596

Average Payload Mass by F9 v1.1

- The average payload mass carried by booster version F9 v1.1 was 2,928.4 kg.

Display average payload mass carried by booster version F9 v1.1

```
%sql SELECT avg("PAYLOAD_MASS__KG_") as avg_payload_mass FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1'
```

```
 * sqlite:///my_data1.db
Done.
```

avg_payload_mass
2928.4

First Successful Ground Landing Date

- The first successful landing outcome on ground pad was achieved on Dec 22, 2015.

List the date when the first succesful landing outcome in ground pad was acheived.

Hint:Use min function

```
%sql SELECT min("Date") as dt_land FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)'
```

```
 * sqlite:///my_data1.db
Done.
```

dt_land
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- Boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000 include: F9 FT B1022, F9 FT B1026, F9 FT B1021.2, and F9 FT B1031.2.

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000 ¶

```
%sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE ("Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS__KG_" > 4000 AND "PAYLOAD_MASS__KG_" < 6000)
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Overall, 100 missions had a successful outcome while one mission had a failure in flight.

List the total number of successful and failure mission outcomes ¶

```
%sql SELECT Mission_Outcome, count(*) as outcome FROM SPACEXTABLE GROUP BY  Mission_Outcome
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	outcome
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- Booster which have carried the maximum payload mass include are F9 B5 variety.

F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- In 2015, there were two drone ship failures. Both occurred at launch site CCAFS LC-40. The first occurred in January with booster version F9 v1.1B1012 and the second occurred in April with booster version F9 v1.1 B1015.

List the records which will display the month names, failure landing_outcomes in drone ship ,booster versions, launch_site for the months in year 2015.

Note: SQLLite does not support monthnames. So you need to use substr(Date, 6,2) as month to get the months and substr(Date,0,5)='2015' for year.

```
%sql SELECT substr(Date, 6,2) as month, Booster_Version, Launch_Site FROM SPACEXTABLE WHERE (substr(Date,0,5)='2015' and Landing_Outcome='Failure (drone ship)')
```

```
 * sqlite:///my_data1.db
Done.
```

month	Booster_Version	Launch_Site
01	F9 v1.1 B1012	CCAFS LC-40
04	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

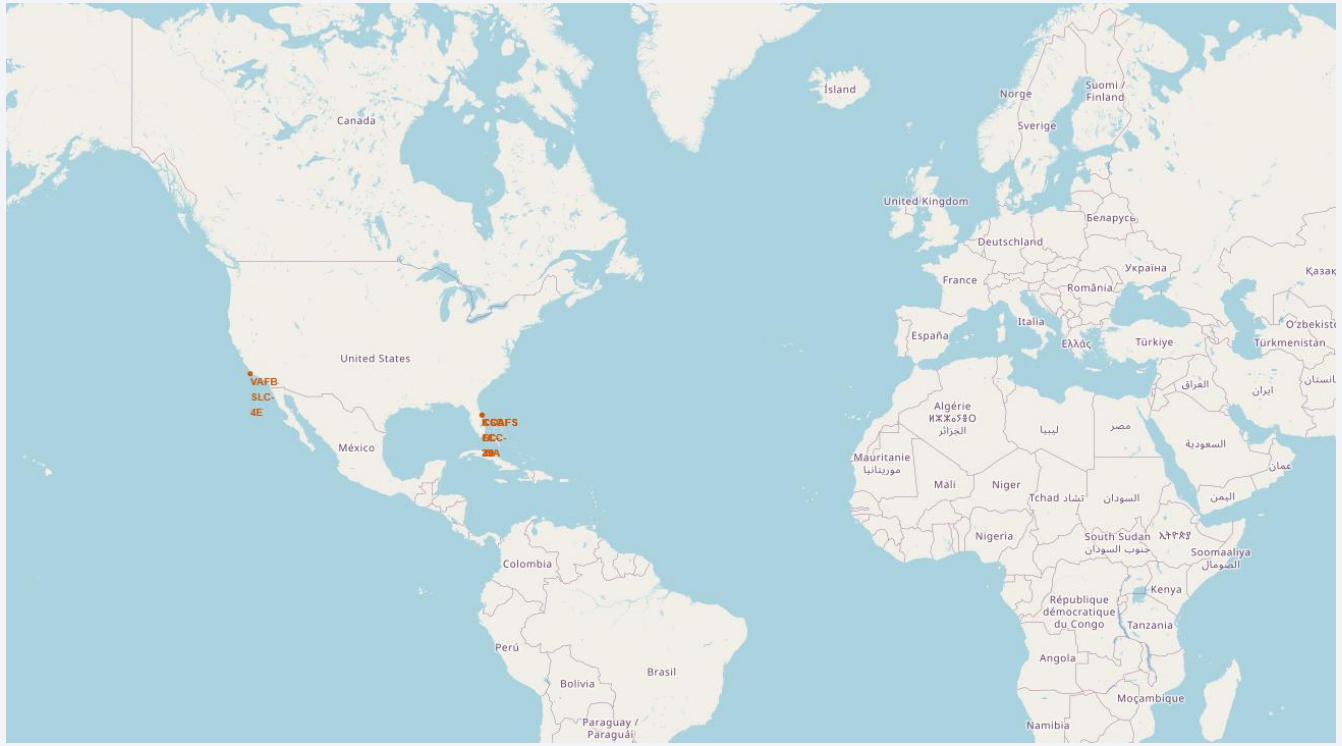
- We see the most common landing outcome was no attempt (n=10), while the least common was precluded (drone ship).

Landing_Outcome	count_outcomes
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

Section 3

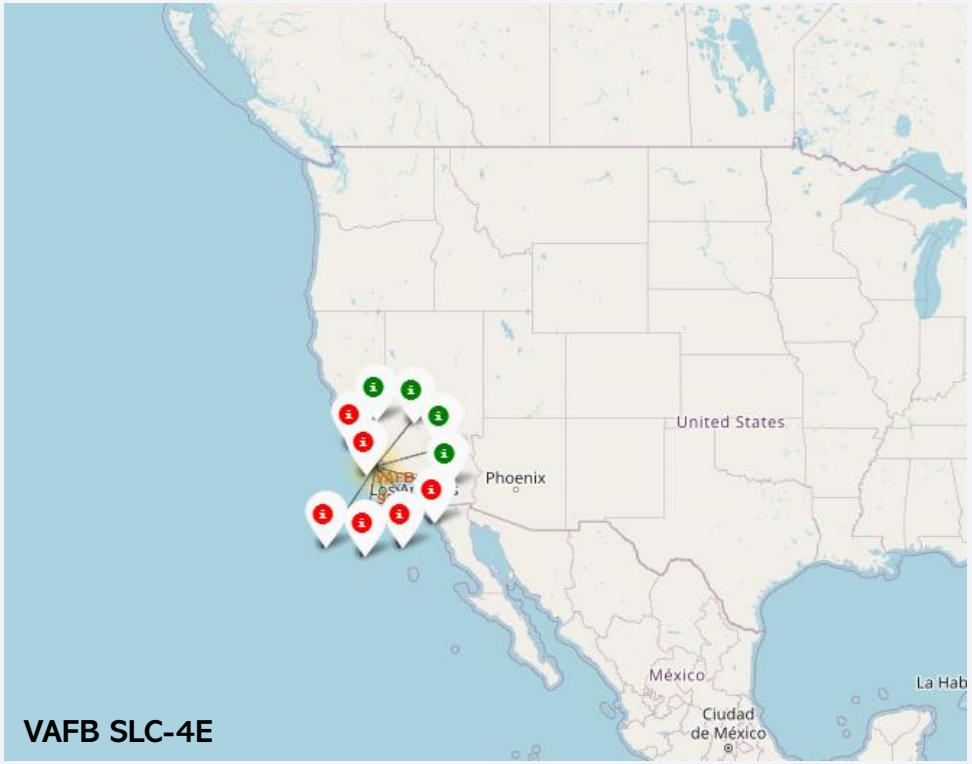
Launch Sites Proximities Analysis

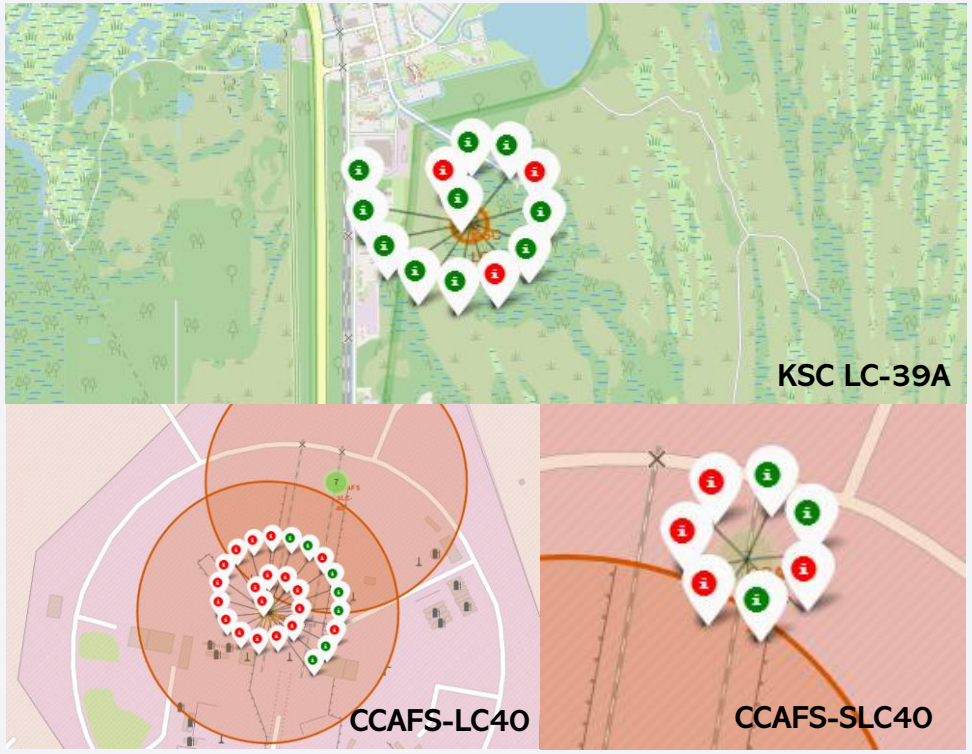
Geographic Spread of SpaceX Falcon 9 Launch Sites



As illustrated on this map, the four locations for SpaceX Falcon 9 launch sites are found within the United States, and three sites are located within Florida, and a fourth is in California.

Successful Landings, By SpaceX Launch Site

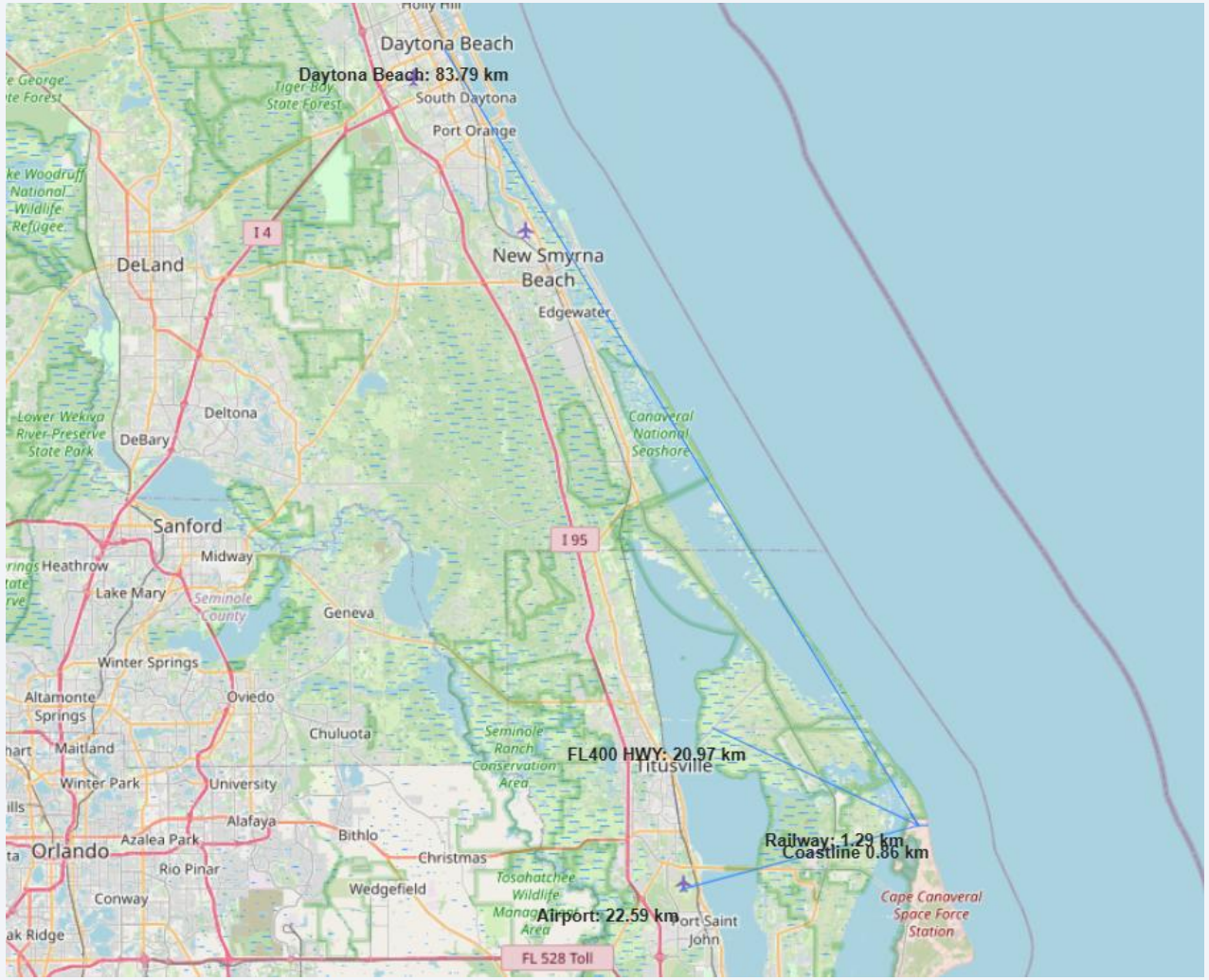




Green markers represent a success, while red markers indicate a failure. From this map we can see the launch site in California (VAFB SLC-4E), had 10 launches and only 4 launches resulted in successful landings of the first stage. Overall KSC LC-39A had the greatest number of successful landings.

Points of Interest, Distance from Launch Site CCAFS SLC-40

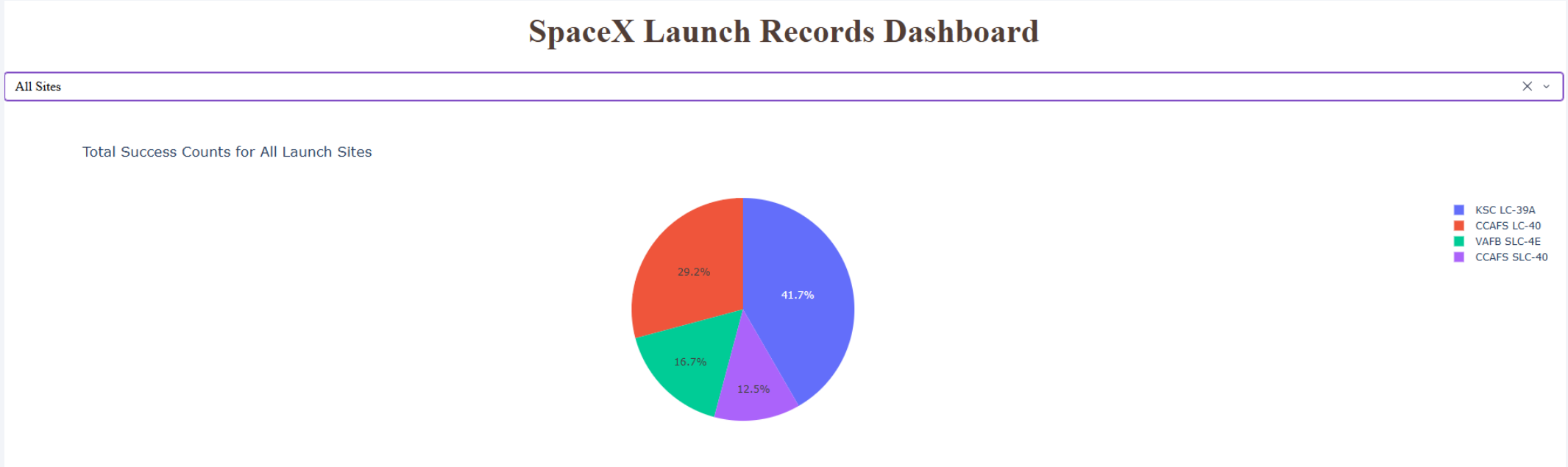
- Launch site CCAFS SLC-40 is located at [28.563197, -80.576820]
- This map shows the distance between launch site CCAFS SLC-40 and points of interest. The launch site is 0.86km west of the Atlantic coastline. A railway is nearby, 1.29km NW, and a highway is 20.97km NW. An airport is located 22.59km SW, while a larger city, Daytona Beach, is located 83.79km NW from the launch site.



Section 4

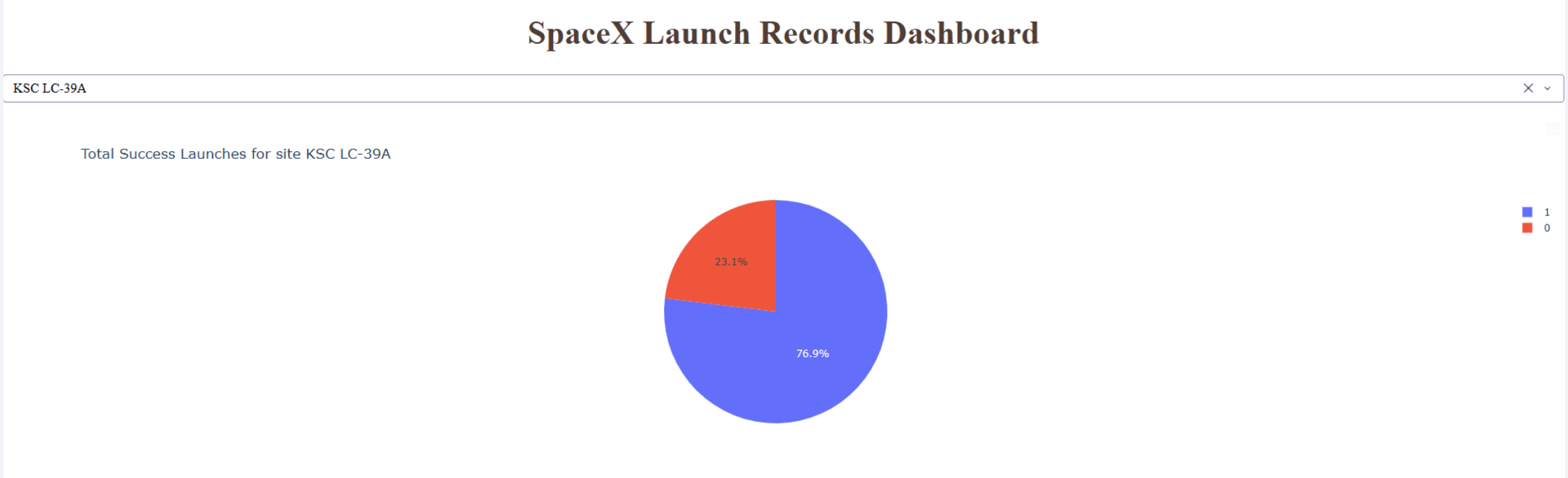
Build a Dashboard with Plotly Dash

Total Success Ratio for All Launch Sites



This pie chart illustrates that among all successful landings, launch site KSC LC-39A contributed the most at 41.7%. In contrast, only 12.5% of total landing successes were from launch site CCAFS SLC-40.

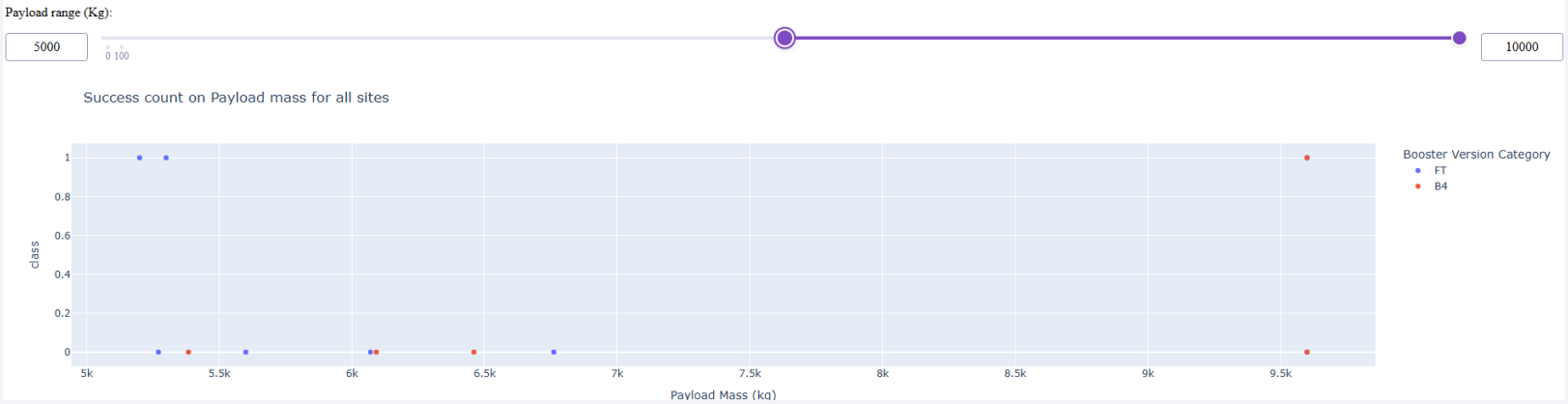
Successes and Failures at Launch Site KSC LC-39A

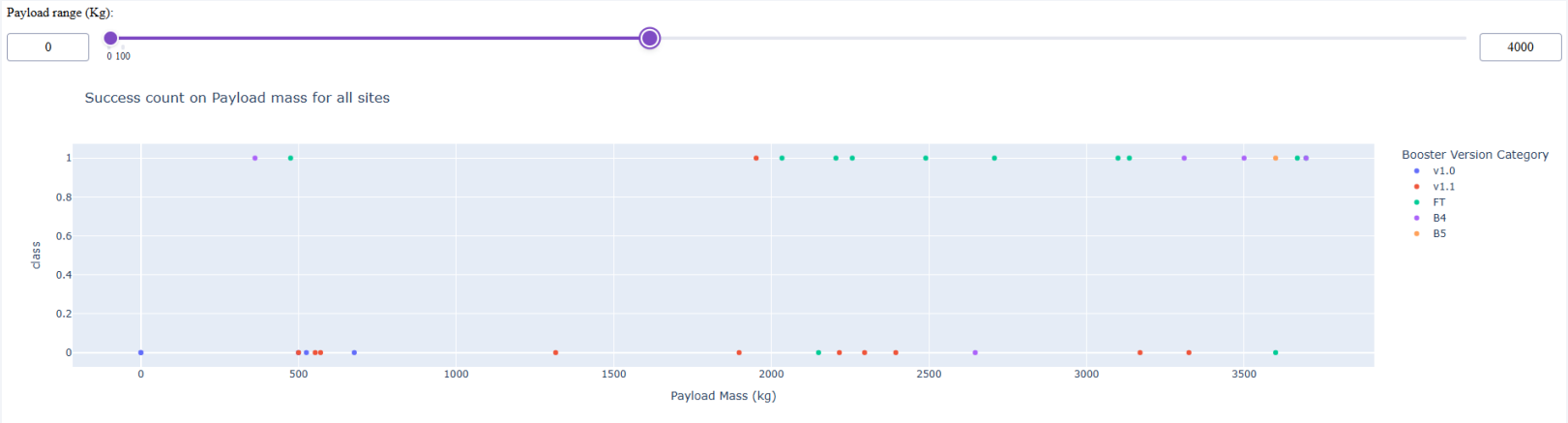


Launch site KSC LC-39A had the highest landing success ratio. Overall, site KSC LC-39A had 76.9% successes.

Payload Mass and Mission Outcome, by Booster Version

- When the payload mass range is 5,000kg or more, only FT and B4 boosters are used, and majority of missions fail.
- When the payload is <5,000kg, a greater variety of boosters are used, and there are more landing successes, with majority appearing to be from the FT booster.



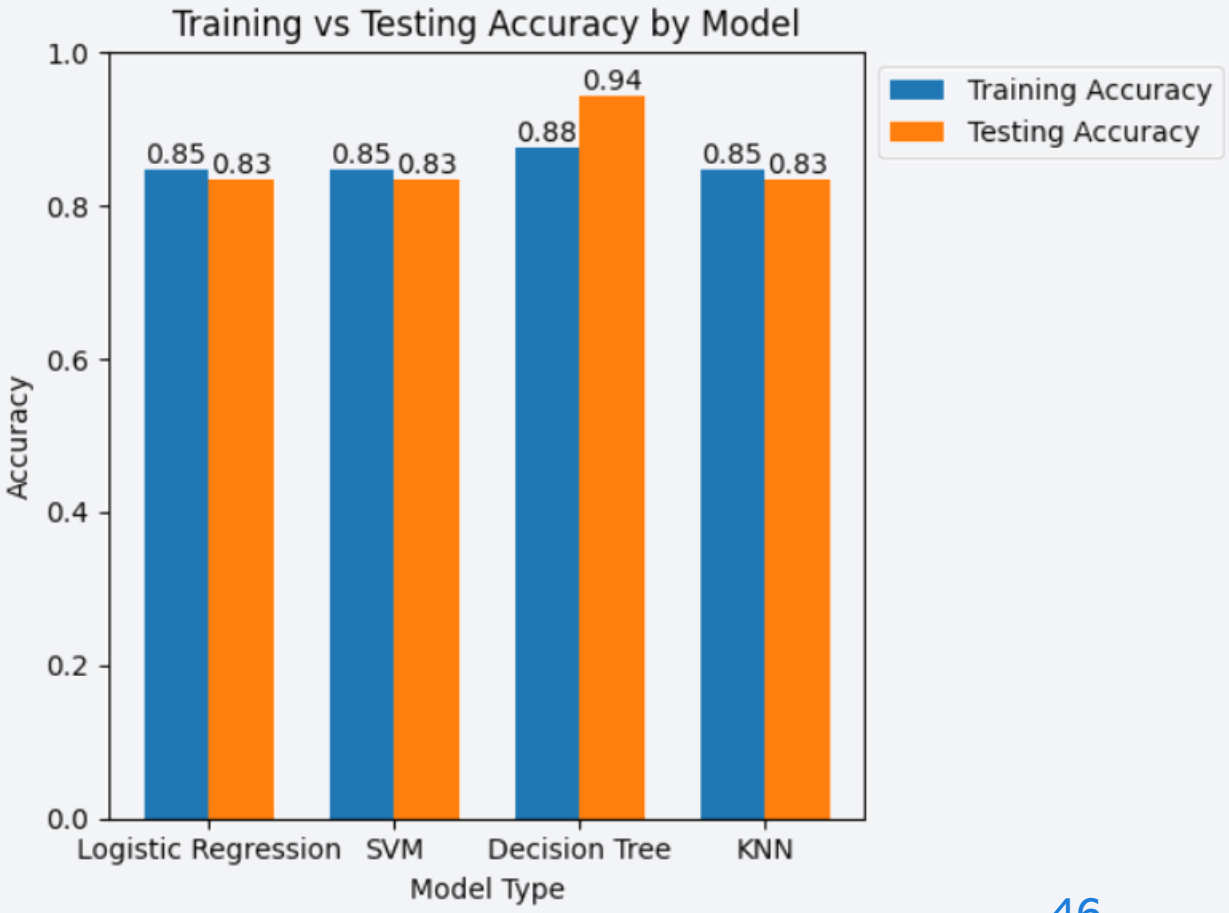


Section 5

Predictive Analysis (Classification)

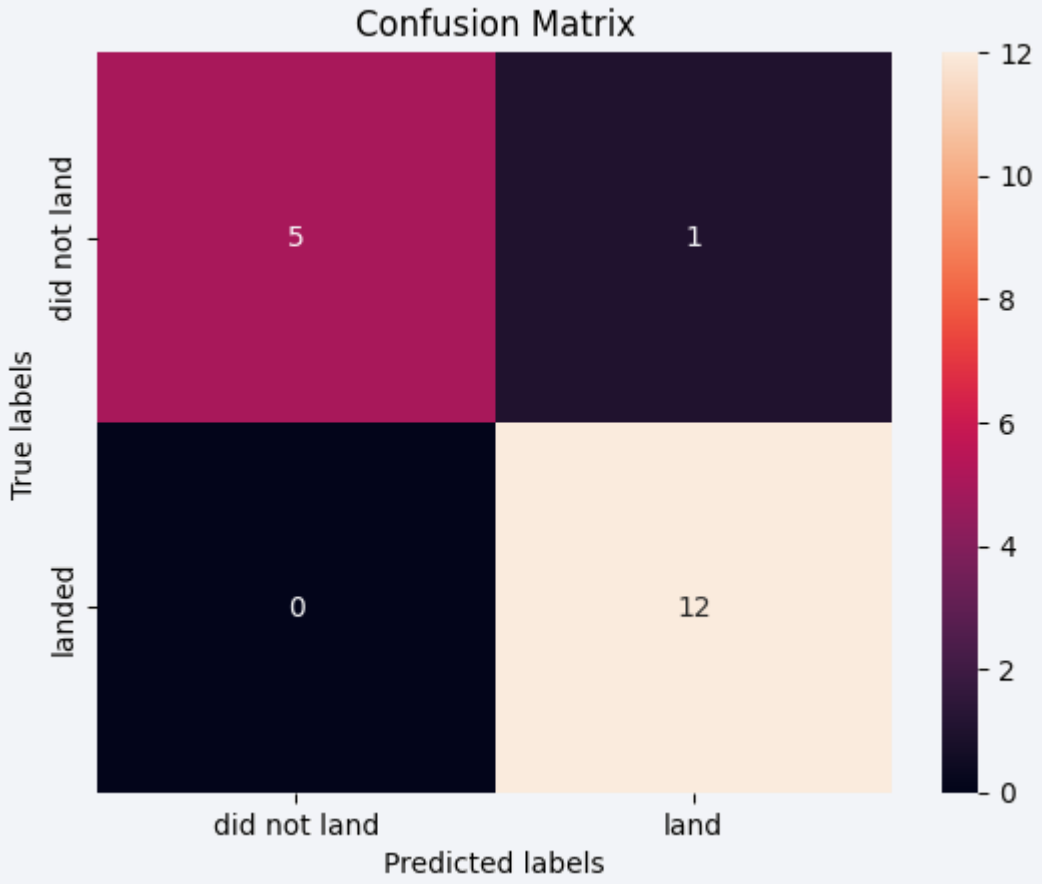
Classification Accuracy

- The bar chart on the right illustrates that the logistic regression, SVM, and KNN models behaved similarly. They each had an accuracy of 0.83 on the testing dataset.
- The decision tree classification model had the highest testing accuracy of 0.94.



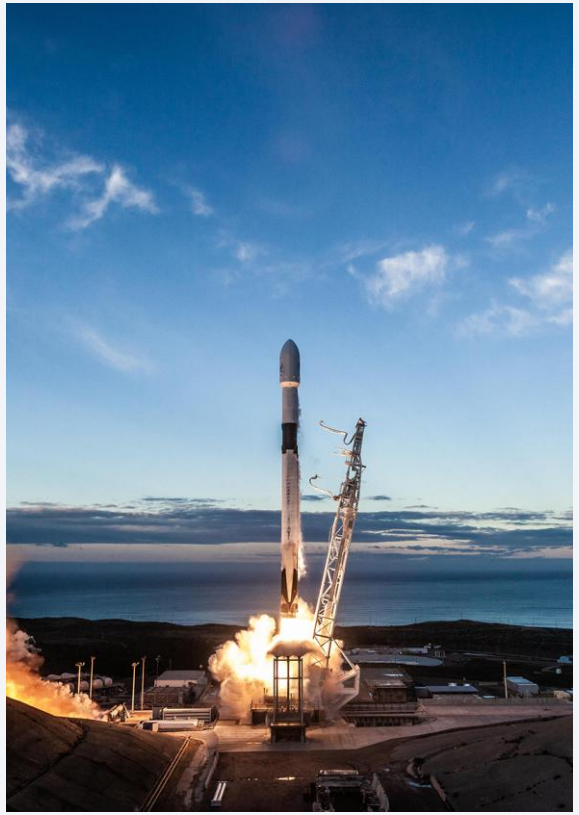
Confusion Matrix

- This is the confusion matrix for the decision tree classifier model. Compared to the other models tested, this model has the fewest false-positive, with a precision of 0.92 and recall of 0.83.



Conclusions

- In this review of SpaceX Falcon 9 data, we observed associations between flight number, payload mass, and launch location.
- Data was transformed and standardized for processing with machine learning algorithms: logistic regression, SVM, decision tree classifier, and KNN.
- We observe the decision tree classifier had the highest accuracy and predicted the fewest false positives.



Innovative Insights

- Overall, this course provided an opportunity to explore the concepts learned throughout the modules and apply these skills to a real-world dataset. However, the methods utilized are not always effective. For example, the plotly dashboard illustrated successes and failures, by booster type and payload. Success and failure is not a continuous variable; therefore a scatterplot was not the most appropriate choice to display these data.
- For the context of this project, we must consider the false positive rate, due to the substantial cost burden. Predicting a successful landing suggests we can reuse the first stage, however a false positive indicates the landing was not actually successful and we will need to pay for a new first stage.
- It is unexpected that the testing model accuracy would be higher than the training model, as seen in the decision tree classifier model. However, this may be the result of our small testing set size, and additional parameters should be explored.

Thank you!